

Han Lin

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Research Interests

Generative models, World models, Multimodal learning, Representation Learning, Reinforcement Learning, Theory-grounded algorithms for efficient Transformers

Education

University of North Carolina at Chapel Hill 2023 - Exp. 2028
Ph.D. in Computer Science GPA: 4.0/4.0

- MURGe-Lab. Advised by Prof. Mohit Bansal

Columbia University 2021 - 2023
M.S. in Computer Science GPA: 4.13/4.33

- DVMM Lab. Advised by Prof. Shih-Fu Chang
- ROAM Lab. Advised by Prof. Matei Ciocarlie and Prof. Shuran Song

Relevant Courses: Learning Theory, Algorithms, Machine Learning, Unsupervised Learning, Bandits & Reinforcement Learning, Causal Inference, Computer Vision, Robotics Learning

Columbia University 2018 - 2020
M.S. in Financial Engineering GPA: 4.04/4.33

- Advised by Prof. Krzysztof Choromanski from Google Deepmind

Relevant Courses: Optimization, Combinatorial Optimization, Stochastic Models, Stochastic Calculus, Monte Carlo Methods, Statistical Inference, Bayesian Statistics, Graphical Models

Central University of Finance and Economics 2014 - 2018
B.S. in Financial Engineering GPA: 90.43/100

Relevant Courses: Linear Algebra, Mathematical Analysis, Probability, Statistics, Real Analysis, Numerical Methods, Stochastic Process, Differential Equations

Preprints

1. **Seeing Isn't Knowing: Do VLMs Know When Not to Answer Spatial Questions (and Why)?**

Yue Zhang, Zun Wang, Han Lin, Jialu Li, Yonatan Bitton, Idan Szpektor, Mohit Bansal
[arxiv | project page]

2. **Planning with Sketch-Guided Verification for Physics-Aware Video Generation**

Yidong Huang, Zun Wang, Han Lin, Dong-Ki Kim, Shayegan Omidshafiei, Jaehong Yoon, Yue Zhang, Mohit Bansal
[arxiv | github]

3. **Training-free guidance in text-to-video generation via multimodal planning and structured noise initialization**

Jialu Li*, Shoubin Yu*, Han Lin*, Jaemin Cho, Jaehong Yoon, Mohit Bansal
[arxiv | github]

Publications

1. **Exploring MLLM-Diffusion Information Transfer with MetaCanvas**
Han Lin, Xichen Pan, Ziqi Huang, Ji Hou, Jialiang Wang, Weifeng Chen, Zecheng He, Felix Junfei-Xu, Junzhe Sun, Zhipeng Fan, Ali Thabet, Jaemin Cho, Mohit Bansal, Chu Wang
NeurIPS 2026 [arxiv | project page]
2. **PhyMotion: Structured 3D Motion Reward for Physics-Grounded Human Video Generation**
Yidong Huang, Zun Wang, Han Lin, Dong-Ki Kim, Shayegan Omidshafiei, Jaehong Yoon, Jaemin Cho, Yue Zhang, Mohit Bansal
NeurIPS 2026 [arxiv | github]
3. **V-Co: A Closer Look at Visual Representation Alignment via Co-Denoising**
Han Lin, Xichen Pan, Zun Wang, Yue Zhang, Chu Wang, Jaemin Cho, Mohit Bansal
ECCV 2026 [arxiv | project page]
4. **AnchorWeave: World-Consistent Video Generation with Retrieved Local Spatial Memories**
Zun Wang, Han Lin, Jaehong Yoon, Jaemin Cho, Yue Zhang, Mohit Bansal
ECCV 2026 [arxiv | project page]
5. **Error-Driven Scene Editing for 3D Grounding in Large Language Models**
Yue Zhang, Zun Wang, Han Lin, Jialu Li, Jianing Yang, Yonatan Bitton, Idan Szepktor, Mohit Bansal
ECCV 2026 [arxiv | github]
6. **EPiC: Efficient Video Camera Control Learning with Precise Anchor-Video Guidance**
Zun Wang, Jaemin Cho, Jialu Li, Han Lin, Jaehong Yoon, Yue Zhang, Mohit Bansal
ICML 2026 [arxiv | project page]
7. **A Benchmark of Expert-Level Academic Questions to Assess AI Capabilities**
Center for AI Safety, Scale AI and HLE Contributors Consortium
Nature, 2026 [paper | project page]
8. **Bifrost-1: Bridging Multimodal LLMs and Diffusion Models with Patch-level CLIP Latents**
Han Lin, Jaemin Cho, Amir Zadeh, Chuan Li, Mohit Bansal
NeurIPS 2025 [arxiv | project page]
9. **Ctrl-Adapter: An Efficient and Versatile Framework for Adapting Diverse Controls to Any Diffusion Model**
Han Lin*, Jaemin Cho*, Abhay Zala, Mohit Bansal
ICLR 2025 (Oral, Top 1.82%) [arxiv | project page]
10. **VEDiT: Latent Prediction Architecture For Procedural Video Representation Learning**
Han Lin, Tushar Nagarajan, Nicolas Ballas, Mido Assran, Mojtaba Komeili, Mohit Bansal, Koustuv Sinha
ICLR 2025 [arxiv]
11. **DreamRunner: Fine-Grained Storytelling Video Generation with Retrieval-Augmented Motion Adaptation**
Zun Wang, Jialu Li, Han Lin, Jaehong Yoon, Mohit Bansal
AAAI 2025 [arxiv | project page]
12. **Fast Tree-Field Integrators: From Low Displacement Rank to Topological Transformers**
Krzysztof Choromanski, Arijit Sehanobish, Somnath Basu Roy Chowdhury, Han Lin, Avinava Dubey, Tamas Sarlos, Snigdha Chaturvedi
NeurIPS 2024 [arxiv]
13. **VideoDirectorGPT: Consistent Multi-scene Video Generation via LLM-Guided Planning**
Han Lin, Abhay Zala, Jaemin Cho, Mohit Bansal
COLM 2024 [arxiv | project page]

14. **EnvGen: Generating and Adapting Environments via LLMs for Training Embodied Agents**
Abhay Zala*, Jaemin Cho*, Han Lin, Jaehong Yoon, Mohit Bansal
COLM 2024 [arxiv | project page]
15. **DiagrammerGPT: Generating Open-Domain, Open-Platform Diagrams via LLM Planning**
Abhay Zala, Han Lin, Jaemin Cho, Mohit Bansal
COLM 2024 [arxiv | project page]
16. **Efficient Graph Field Integrators Meet Point Clouds**
Krzysztof Choromanski*, Arijit Sehanobish*, Han Lin*, Yunfan Zhao*, Eli Berger, Alvin Pan, Tetiana Parshakova, Tianyi Zhang, David Watkins, Valerii Likhoshesterov, Somnath Basu Roy Chowdhury, Avinava Dubey, Deepali Jain, Tamas Sarlos, Snigdha Chaturvedi, Adrian Weller
ICML 2023 [arxiv | github]
17. **Supervised Masked Knowledge Distillation for Few-Shot Transformers**
Han Lin*, Guangxing Han*, Jiawei Ma, Shiyuan Huang, Xudong Lin, Shih-Fu Chang
CVPR 2023 [arxiv | github]
18. **Active Tactile Exploration for 3D Object Recognition**
Jingxi Xu*, Han Lin*, Shuran Song, Matei Ciocarlie
ICRA 2023 [arxiv | project page]
19. **From Block-Toeplitz Matrices to Differential Equations on Graphs: Towards a General Theory for Scalable Masked Transformers**
Krzysztof Choromanski*, Han Lin*, Haoxian Chen*, Tianyi Zhang, Arijit Sehanobish, Valerii Likhoshesterov, Jack Parker-Holder, Tamas Sarlos, Adrian Weller, Thomas Weingarten
ICML 2022 [arxiv | github]
20. **Hybrid Random Features**
Krzysztof Choromanski*, Han Lin*, Haoxian Chen*, Yuanzhe Ma*, Arijit Sehanobish*, Deepali Jain, Michael Ryoo, Jake Varley, Andy Zeng, Valerii Likhoshesterov, Dmitry Kalashnikov, Vikas Sindhwani, Adrian Weller
ICLR 2022 [arxiv | github]
21. **Demystifying Orthogonal Monte Carlo and Beyond**
Han Lin*, Haoxian Chen*, Tianyi Zhang, Clement Laroche, Krzysztof Choromanski
NeurIPS 2020 [arxiv | github]

* Co-First Authors, Equal Contribution

Research Experience

Meta FAIR Lab (DReaM Team) Research Scientist Intern Advised by: Marjan Ghazvininejad, Xiaochuang Han, Mengchen Liu, Luke Zettlemoyer - Self-improvement of prompt rewriter and video generation model - On-policy distillation (OPD) for video generation	2026.5 - now
Meta MSL (Movie Gen Team) Research Scientist Intern Advised by: Chu Wang, Ji Hou, Jialiang Wang, Weifeng Chen, Zecheng He, Felix Xu, Junzhe Sun, Zhipeng Fan, Ali Thabet Exploring VLM-Diffusion Information Transfer with MetaCanvas (NeurIPS 2026)	2025.5 - 2026.5

Meta FAIR Lab (JEPA Team)

2024.5 - 2024.12

Research Scientist Intern

Advised by: Koustuv Sinha, Tushar Nagarajan, Nicolas Ballas, Mido Assran, Mojtaba Komeili

- VEDiT: Latent prediction architecture for procedural video representation learning (ICLR 2025)

Google Research Collaboration on Efficient Graph Networks

2022 - 2024

Advised by: Krzysztof Choromanski, Jack Parker-Holder, Somnath Basu Roy Chowdhury, Avinava Dubey, Tamás Sarlós

- Fast Tree-Field Integrators: From Low Displacement Rank to Topological Transformers (NeurIPS 2024)
- Efficient Graph Field Integrators Meet Point Clouds (ICML 2023)
- From Block-Toeplitz Matrices to Differential Equations on Graphs: Towards a General Theory for Scalable Masked Transformers (ICML 2022)

Google Research Collaboration on Random Feature Algorithms

2019 - 2022

Advised by: Krzysztof Choromanski, Deepali Jain, Michael S. Ryoo, Jake Varley, Andy Zeng, Dmitry Kalashnikov, Vikas Sindhwani

- Hybrid Random Features (ICLR 2022)
- Demystifying Orthogonal Monte Carlo and Beyond (NeurIPS 2020)

MURGe-Lab, UNC

2023 - now

Research Assistant, Advised by Prof. Mohit Bansal

- Text-to-video generation, multimodal learning, and LLMs

DVMM Lab, Columbia University

2022 - 2023

Research Assistant, Advised by Prof. Shih-Fu Chang

- Supervised masked knowledge distillation for few-shot Transformers (CVPR 2023)

ROAM Lab, Columbia University

2022 - 2023

Research Assistant, Advised by Prof. Matei Ciocarlie and Prof. Shuran Song

- Active tactile exploration for 3D object recognition (ICRA 2023)

Cornell, Maryland, Max Planck Pre-doctoral Research School

2022

Industry Experience**China Merchant Securities**

Aug. 2020 - Aug. 2021

Option Market Making Quant Trader, Full Time

- Commodity options and futures trading and daily P&L attribution

Teaching Experience

COMS 4231 Analysis of Algorithms

Fall 2022

COMS 4732 Computer Vision II: Learning

Spring 2022

COMS 4721 Machine Learning for Data Science

Spring 2022

QMSS 5073 Machine Learning for Social Science

Fall 2021

IEOR 4007 Optimization Models & Methods for FE

Fall 2019

IEOR 4418 Transportation Analytics & Logistics

Spring 2019

Awards

CVPR 2023 Travel Award	Jun 2023
CVPR community	
Honorable Mention in Mathematical Context in Modeling	Jan 2018
The Consortium for Mathematics and Its Applications (COMAP)	
Scholarship for Excellence in Organization and Leadership	Apr 2017
Central University of Finance and Economics	
2nd Price for Summer & Winter Social Practice Scholarship	Oct 2015
Central University of Finance and Economics	

Technical Skills

Python, C/C++, MATLAB, R, MySQL, LaTeX
 PyTorch, TensorFlow, Large-Scale TPU/GPU Training

Professional Service

Workshop Organizer: Any-To-Any Multimodal Learning Workshop, CVPR, 2026

Conference Reviewer: ICML 2022-2026, NeurIPS 2022-2026, ICLR 2024-2025, CVPR 2025-2026, ICCV 2025, ECCV 2026

Conference Volunteer: RSS 2022

References

Mohit Bansal

(mbansal@cs.unc.edu)
 Department of Computer Science, UNC Chapel Hill
 Relationship: PhD advisor, director of UNC MURGe-Lab.

Shih-Fu Chang

(sc250@columbia.edu)
 Columbia University
 Relationship: master's research advisor, dean of Columbia Engineering.

Shuran Song

(shuran@stanford.edu)
 Stanford University
 Relationship: master's research advisor, director of REAL Lab at Stanford University.

Chu Wang

(wangchu@meta.com)
 Meta
 Relationship: internship advisor, research scientist at Meta MSL.

Krzysztof Choromanski

(kchoro@google.com)
 Google Deepmind Robotics & Columbia University
 Relationship: master's advisor and research collaborator.

Matei Ciocarlie

(matei.ciocarlie@columbia.edu)
 Columbia University
 Relationship: master's research advisor, director of ROAM Lab at Columbia University.

Koustuv Sinha

(koustuvs@meta.com)
 Meta
 Relationship: internship advisor, research scientist at Meta FAIR Lab.

Marjan Ghazvininejad

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 Relationship: internship advisor, research scientist at Meta FAIR Lab.